

Treatment-Effect Distributions in Difference-in-Differences

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Motivation

- Averages can mask heterogeneity of treatment effects $\tau_i := Y_i(1) - Y_i(0)$.

Nearly homogeneous effects

$$\tau \sim N(0, 0.1^2)$$

$$\mathbb{E}[\tau] = 0, \quad \mathbb{P}(|\tau| > 0.5) \approx 0$$

Offsetting gains and losses

$$\tau \sim \frac{1}{2}N(-1, 0.1^2) + \frac{1}{2}N(1, 0.1^2)$$

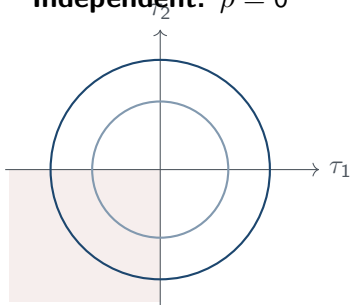
$$\mathbb{E}[\tau] = 0, \quad \mathbb{P}(|\tau| > 0.5) \approx 1$$

- The average cannot reveal the share harmed, the size of losses, or which units gain.
- Those objects matter for incidence, targeting, and welfare.

Motivation

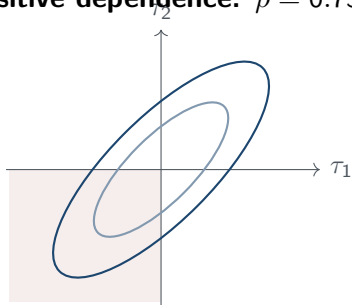
- ▶ In multivariate settings, even identical marginal distributions can imply different joint incidence.
- ▶ Let $\tau \sim N(0, \Sigma)$, where $\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$.

Independent: $\rho = 0$



$$\mathbb{P}(\tau_1 \leq 0, \tau_2 \leq 0) = 25.0\%$$

Positive dependence: $\rho = 0.75$



$$\mathbb{P}(\tau_1 \leq 0, \tau_2 \leq 0) = 38.5\%$$

This paper

► Identification of joint distribution of treatment effects $F_{\tau|D=1}$

- ✓ **Potential outcome equations:** $\Delta \mathbf{Y} = \Delta \mathbf{Y}(0) + D\boldsymbol{\tau}$, $\Delta \mathbf{Y} = (\Delta Y_1, \Delta Y_2)'$.
- ✓ **Two restrictions:** distributional parallel trends and treatment-effect independence,

$$\Delta \mathbf{Y}(0) \mid D = 1 \stackrel{d}{=} \Delta \mathbf{Y}(0) \mid D = 0, \quad \Delta \mathbf{Y}(0) \perp\!\!\!\perp \boldsymbol{\tau} \mid D = 1.$$

- ✓ **Identification:** treated outcome changes are a convolution of untreated changes and treatment effects. Deconvolution identifies the joint distribution of $\boldsymbol{\tau}$.

► Uniform inference

► Empirical application

This paper

- **Identification of joint distribution of treatment effects** $F_{\tau|D=1}$

- **Uniform inference**

- ✓ **Estimation:** estimate the characteristic-function ratio and recover the treatment-effect CDF by known Fourier inversion formulas from the characteristic function

$$\psi_{\tau|D=1}(\lambda) := \frac{\psi_{\Delta Y|D=1}(\lambda)}{\psi_{\Delta Y|D=0}(\lambda)}$$

- ✓ **Linearization:** account for sampling noise from both the treated and control samples.
- ✓ **Uniform inference:** use studentization and a multiplier bootstrap to construct uniform confidence bands for joint distributions.

- **Empirical application**

This paper

- ▶ **Identification of joint distribution of treatment effects** $F_{\tau|D=1}$
- ▶ **Uniform inference**
- ▶ **Empirical application**
 - ✓ **Setting:** displaced workers from mass-layoff who are re-employed one year later in the Veneto Worker History, 1975–2001.
 - ✓ **Outcomes:** annual earnings, daily wages, and AKM wage-premium of the destination employer.
 - ✓ **Question:** do earnings, wage, and employer-premium losses fall on the same workers?

Contributions

- ▶ **Treatment-effect distributions.** Existing work obtains bounds from marginal potential-outcome distributions or copula restrictions. We point identify joint effect distributions under distributional parallel trends + independence between τ and $\Delta Y(0)$.

Heckman, Smith, and Clements (1997); Fan and Park (2009, 2010); Callaway (2021); Callaway and Li (2019).

- ▶ **Deconvolution with an estimated error distribution.** We linearize an estimated characteristic-function ratio so treated and control samples both remain first order, then obtain studentized uniform bands.

Arellano and Bonhomme (2012); Chernozhukov et al. (2014, 2016); Kato and Sasaki (2018); Kurisu and Otsu (2022).

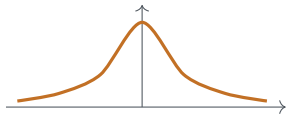
- ▶ **Joint incidence after job displacement.** We study whether wage losses and moves to lower-premium employers fall on the same re-employed workers, rather than only their average contribution to wage losses.

Lachowska, Mas, and Woodbury (2020); Raposo, Portugal, and Carneiro (2021).

Identification and Estimation

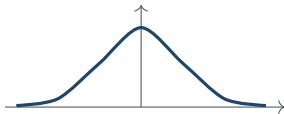
Identification and Deconvolution

Controls: $\Delta \mathbf{Y} \mid D = 0$



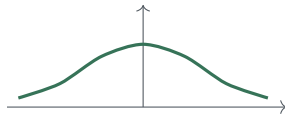
+

Target: τ



=

Treated: $\Delta \mathbf{Y} \mid D = 1$



$$\mathbb{E}[\exp(i\lambda\Delta\mathbf{Y}) \mid D = 1] = \mathbb{E}[\exp(i\lambda[\tau + \Delta\mathbf{Y}(0)]) \mid D = 1]$$

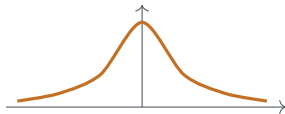
$$\implies \psi_{\Delta\mathbf{Y}|D=1}(\lambda) = \psi_{\tau|D=1}(\lambda)\psi_{\Delta\mathbf{Y}(0)|D=1}(\lambda) \quad \text{by TI: } \tau \perp \Delta\mathbf{Y}(0) \mid D = 1$$

$$\implies \psi_{\Delta\mathbf{Y}|D=1}(\lambda) = \psi_{\tau|D=1}(\lambda)\psi_{\Delta\mathbf{Y}|D=0}(\lambda) \quad \text{by PT: } D \perp \Delta\mathbf{Y}(0)$$

$$\implies \psi_{\tau|D=1}(\lambda) = \frac{\psi_1(\lambda)}{\psi_0(\lambda)}.$$

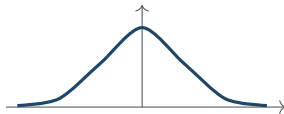
Identification and Deconvolution

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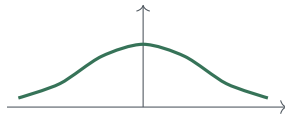
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$$\mathbb{E}[\exp(i\lambda\Delta\mathbf{Y}) \mid D = 1] = \mathbb{E}[\exp(i\lambda[\tau + \Delta\mathbf{Y}(0)]) \mid D = 1]$$

$$\implies \Psi_{\Delta\mathbf{Y}|D=1}(\lambda) = \Psi_{\tau|D=1}(\lambda)\Psi_{\Delta\mathbf{Y}(0)|D=1}(\lambda) \quad \text{by TI: } \tau \perp \Delta\mathbf{Y}(0) \mid D = 1$$

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$$\implies \Psi_{\tau|D=1}(\lambda) = \frac{\Psi_1(\lambda)}{\Psi_0(\lambda)}.$$

It resembles the standard (mean) case in logs: $\log \Psi_{\tau|D=1} = \log \Psi_1 - \log \Psi_0$.

Intuition: Untreated changes do not predict treatment effects among the treated.

Deconvolution (Inversion) Formulas

For coordinate j , Gil-Pelaez (1951) yields the marginal CDFs

$$F_{\tau_j|D=1}(z) = \frac{1}{2} - \frac{1}{\pi} \int_0^\infty \operatorname{Im} \left\{ \frac{e^{-i\omega z}}{\omega} \frac{\Psi_{1,j}(\omega)}{\Psi_{0,j}(\omega)} \right\} d\omega,$$
$$F_{h,\tau_j|D=1}(z) = \frac{1}{2} - \frac{1}{\pi} \int_0^\infty \operatorname{Im} \left\{ \frac{e^{-i\omega z}}{\omega} \frac{\Psi_{1,j}(\omega)}{\Psi_{0,j}(\omega)} \Psi_K(h\omega) \right\} d\omega,$$

where $\Psi_K(h\omega)$ is the characteristic function of a kernel function, and supported in $[-1, 1]$.

Similarly, Shephard (1991) yields the bivariate CDF

$$F_{\tau|D=1}(\mathbf{z}) = \frac{F_{\tau_1|D=1}(z_1) + F_{\tau_2|D=1}(z_2)}{2} - \frac{1}{4}$$
$$- \frac{1}{2\pi^2} \int_0^\infty \int_0^\infty \nabla_{\lambda_2} \operatorname{Re} \left\{ \frac{e^{-i\lambda'z}}{\lambda_1 \lambda_2} \frac{\Psi_1(\lambda)}{\Psi_0(\lambda)} \right\} d\lambda_1 d\lambda_2,$$

where $F_{h,\tau|D=1}(\mathbf{z})$ multiplies $\Psi_K(\lambda h) := \Psi_K(\lambda_1 h) \Psi_K(\lambda_2 h)$.

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where $F_{h,\tau|D=1}(\mathbf{z})$ multiplies $\Psi_K(\lambda h) := \Psi_K(\lambda_1 h) \Psi_K(\lambda_2 h)$.

Estimators use the plug-in estimators of the characteristic functions: $\hat{\Psi}_1$, and $\hat{\Psi}_0$

Uniform Inference of $F_{\tau|D=1}(\mathbf{z})$

(over a compact set $\mathbf{z} \in \mathcal{I}$)

Asymptotic Linear Representation

Let $\delta_{d,n} = \widehat{\Psi}_d - \Psi_d$. The exact ratio identity is

$$\frac{\widehat{\Psi}_1}{\widehat{\Psi}_0} - \frac{\Psi_1}{\Psi_0} = \underbrace{\frac{\delta_{1,n}}{\Psi_0}}_{\text{Numerator noise}} - \underbrace{\frac{\Psi_\tau}{\Psi_0} \delta_{0,n}}_{\text{Denominator noise}} - \underbrace{\frac{\delta_{1,n} \delta_{0,n}}{\Psi_0 \widehat{\Psi}_0} + \frac{\Psi_\tau \delta_{0,n}^2}{\Psi_0 \widehat{\Psi}_0}}_{\text{Quadratic remainder}}.$$

Our main theory assumes $|\Psi_0(\boldsymbol{\lambda})| \sim (1 + \|\boldsymbol{\lambda}\|)^{-\beta}$, and the remainder is of order $O_p(n^{-1} h^{-2\beta} \log h^{-1})$ on $\|\boldsymbol{\lambda}\|_\infty \leq h^{-1}$.

This uniform rate is not enough due to the integral and singularity at $\boldsymbol{\lambda} = 0$. We show that

$$\widehat{F}_{\tau|D=1}(\mathbf{z}) - F_{h,\tau|D=1}(\mathbf{z}) = \frac{1}{n} \sum_{i=1}^n \psi_{\mathbf{z}}(W_i) + R_n(\mathbf{z}),$$

$$\sup_{\mathbf{z} \in \mathcal{I}} |R_n(\mathbf{z})| = O_p(n^{-1} h^{-2\beta} (\log h^{-1})^3)$$

Constructing Confidence Bands

Consider the following Studentized process:

$$\hat{Z}_n(\mathbf{z}) = \frac{\sqrt{n}\{\hat{F}_{\tau|D=1}(\mathbf{z}) - F_{h,\tau|D=1}(\mathbf{z})\}}{\hat{\sigma}_n(\mathbf{z})}.$$

Gaussian coupling (Theorem 2)

$$\sup_t \left| \mathbb{P}(\|\hat{Z}_n\|_{\mathcal{I}} \leq t) - \mathbb{P}(\|Z_n^G\|_{\mathcal{I}} \leq t) \right| \rightarrow 0,$$

where $\|Z_n^G\|_{\mathcal{I}} = O_p(\sqrt{\log h^{-1}})$.

If $\hat{c}_n(1 - \alpha)$ is the conditional quantile of $\|\hat{Z}_n^\xi\|_{\mathcal{I}}$, then

$$\hat{C}_n(\mathbf{z}) = \left[\hat{F}_{\tau|D=1}(\mathbf{z}) \pm \frac{\hat{\sigma}_n(\mathbf{z})}{\sqrt{n}} \hat{c}_n(1 - \alpha) \right].$$

Multiplier Bootstrap (Theorem 3)

$$\hat{Z}_n^\xi(\mathbf{z}) = \frac{1}{\hat{\sigma}_n(\mathbf{z})\sqrt{n}} \sum_i \xi_i \{\hat{\psi}_{\mathbf{z}}(W_i) - \bar{\psi}_{\mathbf{z}}\}.$$

Conditionally on the data, its supremum couples to the same Gaussian supremum.

The coupling covers F_h ; undersmoothing makes $F_h - F$ negligible at the scale $\frac{\sqrt{n}}{\inf_{\mathbf{z} \in \mathcal{I}} \sigma(\mathbf{z})} \sqrt{\log h^{-1}}$.

Simulation Results

- ▶ $\Delta \mathbf{Y} = \Delta \mathbf{Y}(0) + D\boldsymbol{\tau}$, $\mathbb{P}(D = 1) = \frac{1}{2}$.
- ▶ We simulate $\Psi_{\Delta \mathbf{Y}(0)}(\boldsymbol{\lambda}) = (1 + \frac{1}{2}\boldsymbol{\lambda}'\Sigma_{\Delta}\boldsymbol{\lambda})^{-1}$.
 - ✓ This is a multivariate Laplace distribution. Satisfies ordinary-smooth design with $\beta = 2$.
- ▶ Treatment effects follow $\boldsymbol{\tau} \sim N(\mathbf{0}, \Sigma_{\tau})$, $\rho_{\tau} = 0.5$.
- ▶ We let $\rho_{\varepsilon} \in \{0, 0.25\}$.
- ▶ Define the signal-to-noise ratio as $\sigma_{\tau}/\sigma_{\varepsilon} \in \{2, 4, 8\}$.
 - ✓ A smaller ratio makes harder to disentangle treatment effects from the controls' distribution in finite samples.

Simulation Results

Nominal Level	n	Coverage			Average band length		
		SNR 2	SNR 4	SNR 8	SNR 2	SNR 4	SNR 8
<i>Panel A. $\rho_\varepsilon = 0$</i>							
0.80	1,000	.819	.797	.802	.155	.098	.101
	2,000	.839	.816	.796	.125	.072	.072
0.90	1,000	.908	.898	.898	.171	.111	.114
	2,000	.923	.908	.902	.138	.081	.081
0.95	1,000	.961	.944	.945	.185	.122	.124
	2,000	.955	.942	.955	.149	.089	.089
<i>Panel B. $\rho_\varepsilon = 0.25$</i>							
0.80	1,000	.805	.776	.812	.156	.099	.101
	2,000	.824	.804	.795	.126	.073	.072
0.90	1,000	.912	.897	.901	.172	.112	.113
	2,000	.919	.902	.905	.139	.082	.081
0.95	1,000	.958	.942	.951	.187	.123	.124
	2,000	.961	.953	.962	.150	.090	.089

1,000 Monte Carlo replications; 1,000 multiplier draws in each replication.

Empirical Application:

Unequal Consequences of Job Displacement

Job Displacement in Veneto, Italy

► Data Source

- ✓ Veneto Workers History matched employer–employee panel, 1975–2001.
- ✓ Keep employment spell with the highest annual earnings (main job).

► DiD Design

► Sample and Functionals of Interest

Job Displacement in Veneto, Italy

► Data Source

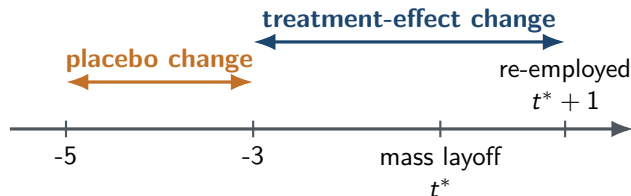
► DiD Design

- ✓ Treated displacement at t^* : separation from the main employer when employer employment falls by at least 30%, including closure.
- ✓ Match to untreated workers by calendar year, gender, and industry cells.
- ✓ One-to-one nearest-neighbor matching without replacement. Propensity score uses earnings at $t^* - 2$, $t^* - 3$, age, tenure, and employer size at $t^* - 1$.
- ✓ Outcomes of interest: log earnings, log daily wages and AKM wage fixed effects.
- ✓ **PT**: earnings/wages/AKM FE changes distribution would have been identical to the matched controls in the absence of job displacement (given X).
- ✓ **TI**: untreated earnings/wages/AKM FE cannot predict who gains or loses from job displacement (given X).

► Sample and Functionals of Interest

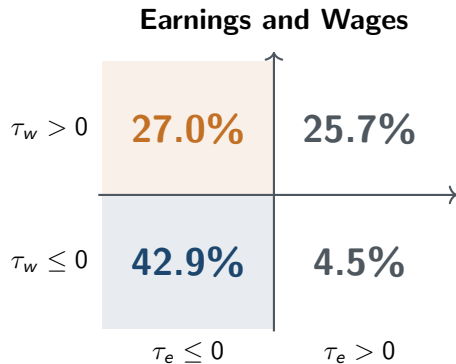
Job Displacement in Veneto, Italy

- ▶ Data Source
- ▶ DiD Design
- ▶ Sample and Functionals of Interest
 - ✓ Sample:



- ✓ We focus on τ as treatment effects on (earnings, wages), and (wages, AKM premia) separately.
- ✓ We care about joint incidence $\mathbb{P}(\tau_1 \leq z_1 \mid \tau_2 \leq z_2)$ and Spearman rank correlation ρ_S between τ .

Results I: Earnings and Wages



Marginal effects

$\mathbb{E}[\tau_e] = -0.193$, while $\mathbb{E}[\tau_w] = -0.002$.

69.9% have a nonpositive annual-earnings effect.

Dependence

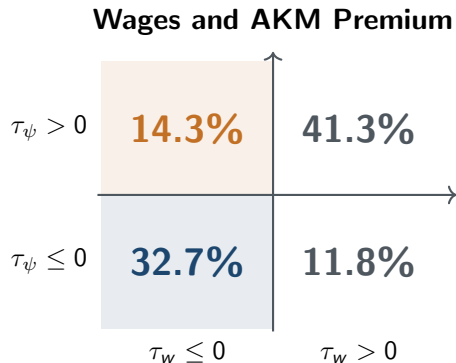
$$\rho_S(\tau_e, \tau_w) = -0.002$$

95% projection interval: $[-0.167, 0.162]$.

The dispersion in annual earnings is consistent with adjustment through days worked, hours, overtime, or job stability—not only daily pay.

► Full functionals

Results II: Wages and AKM premium



Conditional loss incidence

$$\mathbb{P}(\tau_\psi \leq 0 \mid \tau_w \leq 0) = 69.6\%$$

$$\mathbb{P}(\tau_w \leq 0 \mid \tau_\psi \leq 0) = 73.5\%$$

Rank dependence

$$\rho_S(\tau_w, \tau_\psi) = 0.422$$

95% projection interval: [0.305, 0.538].

This supports evidence about employer reallocation as driver to wage impacts.

► AKM construction

Discussion

Discussion

- ▶ We develop a framework to point-identify the Distribution of Treatment Effects (DTE) on multiple dimensions.
- ▶ Inference on functionals of the joint distribution follows from Gaussian coupling and multiplier bootstrap (Chernozhukov et al., 2014; 2016).
- ▶ **Empirical application.** Earnings and wage losses are weakly bundled, while wage losses are more closely associated with moves to “lower-quality” employers.
- ▶ The framework allows to uncover more stylized facts on jobs displacement:
 - ✓ Conditional effects
 - ✓ Testing mechanisms a la Kwon-Roth (2026)
 - ✓ Persistence of treatment effects

▶ Conditional Effects

▶ Testing Mechanisms

▶ Persistence

Thank you!

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Appendix

Identification assumptions in full

For each x in the treated support:

1. **Overlap:** both $D = 1$ and $D = 0$ occur with positive probability conditional on $X = x$.
2. **Distributional parallel trends:**

$$\Delta \mathbf{Y}(0) \mid D = 1, X = x \stackrel{d}{=} \Delta \mathbf{Y}(0) \mid D = 0, X = x.$$

3. **Treatment-effect independence:**

$$\Delta \mathbf{Y}(0) \perp\!\!\!\perp \boldsymbol{\tau} \mid D = 1, X = x.$$

4. **Nonvanishing control transform:** $\Psi_{\Delta \mathbf{Y} \mid D=0, X=x}(\boldsymbol{\lambda}) \neq 0$ for every $\boldsymbol{\lambda} \in \mathbb{R}^2$.
5. **Continuity:** each coordinate of $\boldsymbol{\tau} \mid D = 1, X = x$ has a continuous cdf.

Both groups enter the influence function

Suppress conditioning and let $p = \mathbb{P}(D = 1)$. The joint-transform score has treated and control components:

$$\begin{aligned}\ell_{i,T}^{12}(\lambda) &= \frac{D_i}{p} \frac{e^{i\lambda' \Delta \mathbf{Y}_i} - \psi_1(\lambda)}{\psi_0(\lambda)}, \\ \ell_{i,C}^{12}(\lambda) &= -\frac{1 - D_i}{1 - p} \frac{\psi_\tau(\lambda)}{\psi_0(\lambda)} \left\{ e^{i\lambda' \Delta \mathbf{Y}_i} - \psi_0(\lambda) \right\}.\end{aligned}$$

The influence function $\psi_{\mathbf{z}}(W_i)$ applies the regularized joint and marginal inversion operators to these transform scores. Therefore the same treated/control decomposition carries into

$$\widehat{F}_{\tau|D=1}(\mathbf{z}) - F_{h,\tau|D=1}(\mathbf{z}) = n^{-1} \sum_i \psi_{\mathbf{z}}(W_i) + R_n(\mathbf{z}).$$

Bootstrap target and smoothing bias

The multiplier process is

$$\hat{Z}_n^\xi(\mathbf{z}) = \frac{1}{\hat{\sigma}_n(\mathbf{z})\sqrt{n}} \sum_{i=1}^n \xi_i \left\{ \hat{\psi}_{\mathbf{z}}(W_i) - n^{-1} \sum_{l=1}^n \hat{\psi}_{\mathbf{z}}(W_l) \right\}.$$

The multiplier theorem yields a uniform band around the regularized population target $F_{h,\tau|D=1}$. Coverage of the true cdf additionally requires

$$\frac{\sqrt{n} \|F_{h,\tau|D=1} - F_{\tau|D=1}\|_{\mathcal{I}}}{\inf_{\mathbf{z} \in \mathcal{I}} \sigma_n(\mathbf{z})} \sqrt{\log h^{-1}} \rightarrow 0.$$

◀ Back to talk

Bandwidth Selection

Start with a normal-reference pilot and set $h_0 = 1.5h_{\text{pilot}}$. Form

$$h_j = \frac{j}{J} h_0, \quad j = 1, \dots, J, \quad d_{j-1,j} = \|\widehat{F}_{\tau}^{(h_{j-1})} - \widehat{F}_{\tau}^{(h_j)}\|_{\mathcal{I}}.$$

Move from h_j toward smaller bandwidths. At the first crossing

$$d_{j-1,j} > \rho \|\widehat{F}_{\tau}^{(h_{j-1})} - \widehat{F}_{\tau}^{(h_j)}\|_{\mathcal{I}},$$

select h_j , the last stable candidate before higher-frequency variation changes the estimate materially. The implementation uses

$$J = \lceil 4 \log n \rceil, \quad \rho = 0.4 \log n.$$

Scope

This L^∞ -stability heuristic follows Bissantz et al. (2007) and Kato and Sasaki (2018). The current coverage theorem is for prespecified undersmoothing; it does not prove uniform validity for the random selector.

The matched re-employed sample is balanced on reported observables

	Treated		Control	
	Mean	SD	Mean	SD
Earnings at $t^* - 3$, EUR thousands	40.1	24.8	39.3	25.4
Age	40.9	8.1	41.0	8.3
Female	0.32	0.47	0.32	0.46
Tenure	5.48	2.69	4.96	2.45
Employer size	497	957	501	1,028
Workers, thousands	36.7		36.7	
Firms, thousands	3.3		3.0	

Characteristics measured at $t^* - 1$, except earnings at $t^* - 3$; earnings in 2010 thousand euros.

◀ Back to talk

The AKM outcome measures employer reallocation

$$W_{it} = \alpha_i + \psi_{J(i,t)} + \lambda_t + X'_{it}\gamma + u_{it}.$$

1. Estimate worker and firm effects using pre-policy data.
2. Exclude the matched analysis pairs from the AKM estimation sample.
3. Assign $\hat{\psi}_{J(i,t)}$ to every analysis worker's observed employer.
4. At $t^* + 1$, τ_ψ is the displacement-induced change in the pre-policy premium assigned through the destination employer.

◀ Back to talk

Diagnostics are descriptive finite-grid stress tests

	Earnings–wage	Wage–premium
Bochner discrepancy: descriptive multiplier tail area	0.107	0.211
Treated share in cells flagged at 5%	0.058	0.058
Pre-period discrepancy: descriptive multiplier tail area	0.422	0.842
Treated share in cells flagged at 5%	0.234	0.100

- ▶ The placebo compares the distribution of changes from $t^* - 5$ to $t^* - 3$.
- ▶ The Bochner object asks whether the estimated post-treatment ratio behaves like a characteristic function on the retained grid.

◀ Back to talk

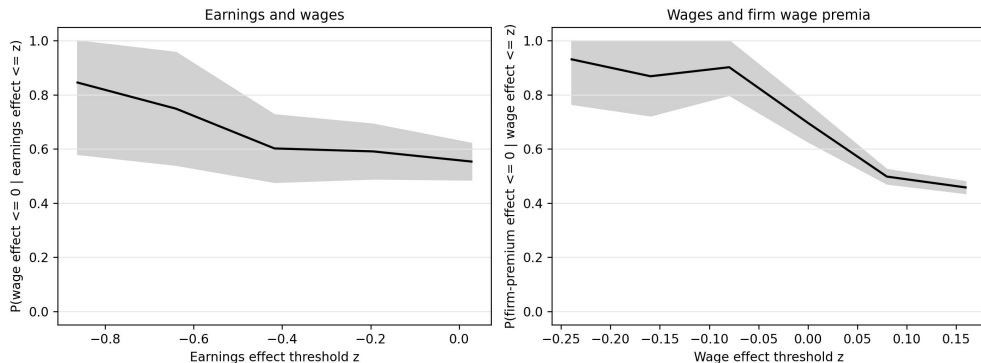
Current marginal and joint functionals

Functional	Earnings–wage	Wage–premium
$\mathbb{E}[\tau_1], \mathbb{E}[\tau_2]$	-0.193, -0.002	-0.002, 0.006
$\text{Var}(\tau_1), \text{Var}(\tau_2)$	0.145, 0.025	0.025, 0.017
$\mathbb{P}(\tau_1 \leq 0, \tau_2 \leq 0)$	0.429 [0.376, 0.482]	0.327 [0.301, 0.353]
$\mathbb{P}(\tau_1 \leq 0, \tau_2 > 0)$	0.270 [0.223, 0.317]	0.143 [0.110, 0.175]
$\mathbb{P}(\tau_2 \leq 0 \mid \tau_1 \leq 0)$	0.614 [0.510, 0.718]	0.696 [0.597, 0.795]
$\mathbb{P}(\tau_1 \leq 0 \mid \tau_2 \leq 0)$	0.906 [0.772, 1.000]	0.735 [0.625, 0.846]
$\rho_S(\tau_1, \tau_2)$	-0.002 [-0.167, 0.162]	0.422 [0.305, 0.538]

Within each pair, $(\tau_1, \tau_2) = (\tau_e, \tau_w)$ or (τ_w, τ_ψ) . Brackets are 95% projection intervals from the uniform cdf band.

◀ Back to talk

Conditional loss curves

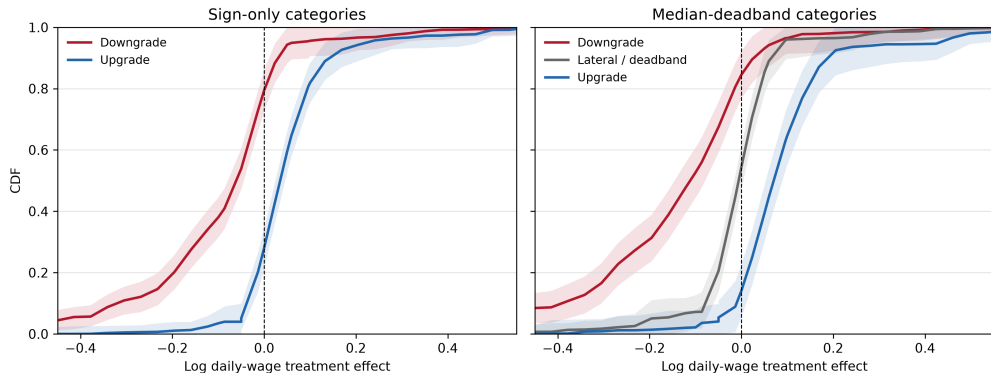


Left: $\mathbb{P}(\tau_w \leq 0 \mid \tau_e \leq z)$. Right: $\mathbb{P}(\tau_\psi \leq 0 \mid \tau_w \leq z)$. The firm premium is the assigned pre-policy destination-firm premium.

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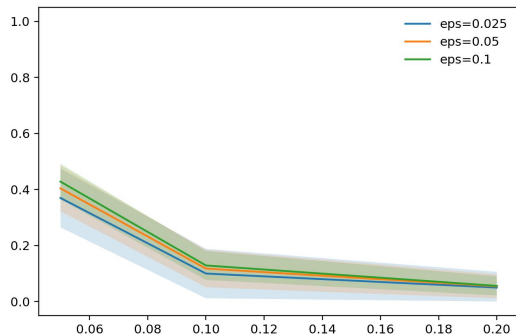
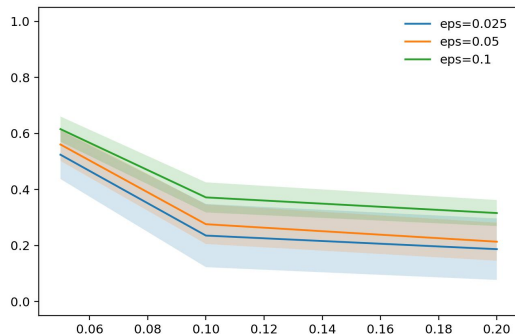
Conditional CDFs

Wage-effect distributions by observed employer-score reallocation at +1



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Mechanism-style curves are reduced-form incidence summaries



Left: $\mathbb{P}(\tau_e \leq z \mid |\tau_w| \leq \epsilon)$. Right: $\mathbb{P}(\tau_w \leq z \mid |\tau_\psi| \leq \epsilon)$. The firm premium is the assigned pre-policy destination-firm premium.

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Persistence of Wage Effects

